

A Novel IoT-Integrated Approach to Waste Management in Residential Complexes

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Abstract—This paper presents the design and implementation of an IoT-enabled waste management system aimed at addressing the issue of excessive organic waste generation in residential buildings. The proposed system leverages an ESP32 microcontroller integrated with an ultrasonic sensor, an MQ135 gas sensor, and an RFID module to enable real-time monitoring and user-specific waste tracking. Residents are required to scan their RFID cards before disposing of organic waste, ensuring accountable usage and individualized data logging. Sensor readings are transmitted to a Firebase real-time database and visualized through a Flutter-based mobile application, which displays waste levels and gas emissions for each user. Threshold-based data handling mechanisms are implemented to detect overuse or potential hazards. By providing a transparent and efficient monitoring platform, the system supports local councils or building management in reducing operational costs while promoting environmentally responsible waste disposal practices. This implementation demonstrates a scalable approach to smart waste management using low-cost IoT components and mobile technologies.

Keywords—component, formatting, style, styling, insert (key words)

I. INTRODUCTION

Organic waste management in Malaysian residential buildings faces significant challenges due to poor tracking and disposal control. These inefficiencies contribute to environmental degradation and operational burdens for building management. As urbanization accelerates, the demand for innovative and sustainable waste management solutions continues to rise. A study in Puncak Alam, Kuala Selangor, reported average household waste generation at 1.0 kg per capita per day, with each household producing about 2.5 kg daily. Organic waste made up 60% of this total. (Yatim et al., 2019) This highlights the urgent need for more active management strategies to reduce environmental impact.

Nationally, waste generation rose from 19,000 tons daily in 2005 to 38,000 tons in 2018, while the recycling rate remained low at 17.5%. (Malaysian Investment Development Authority [MIDA], n.d.) These figures illustrate the increasing strain on Malaysia's waste management systems. To address these challenges, IoT offers a promising solution by enabling real-time tracking through sensors and connected networks. It supports a more efficient, transparent, and environmentally responsible waste management process. (Bridgera, n.d.)

This project proposes an IoT-enabled system for managing organic waste in residential buildings, using sensors, RFID technology, and a mobile application for real-time data collection and monitoring. The system aims to improve operational efficiency and reduce environmental impact.

A. Problem Statement

Many residential buildings today lack structured systems for managing organic waste, leading to unregulated disposal practices. Residents often discard waste without restrictions, increasing management costs and fostering unsustainable habits. This neglects the importance of responsible waste management.

Organic waste contributes significantly to environmental problems such as methane emissions and resource wastage. Moreover, unmanaged waste can cause health hazards, foul odors, and pest infestations, deteriorating the quality of life in residential communities. Without effective tracking and control mechanisms, these challenges persist, threatening environmental sustainability and community well-being. (Climatesort, n.d.)

To address these concerns, this project proposes an IoT-enabled waste management system designed for residential buildings. The system integrates sensors, RFID modules, and mobile applications to enable real-time data collection, continuous monitoring, and instant notifications. This solution provides building management with insights into residents' disposal habits, bin fill levels, and environmental hazards like methane emissions.

B. Research Objective

- Monitor waste volume and gas concentration using ultrasonic and MQ135 sensors.
- Implement RFID-based user identification to promote accountability in waste disposal.
- Provide real-time data visualization through a Flutter mobile application.
- Store and manage sensor data using Firebase for effective monitoring and analysis.

C. Scope and Limitations of the Study

This project focuses on the development of a smart waste management system utilizing Internet of Things (IoT) technology to improve organic waste monitoring in residential buildings. The system integrates ultrasonic sensors and MQ135 gas sensors to measure bin fill levels and detect methane concentrations. RFID technology is employed to log individual residents' waste disposal activities, fostering accountability and usage tracking. An ESP32 microcontroller serves as the core processing unit, collecting sensor data and transmitting it over Wi-Fi to a Firebase cloud database. A Flutter-based mobile application displays real-time data

visualizations, including waste levels and gas concentration readings. The system includes threshold-based alerts, ensuring timely identification of full bins or hazardous gas levels.

The scope is limited to prototype testing within a controlled environment using the actual mobile application. The system does not support large-scale deployment, integration with municipal waste systems, or advanced machine learning-based analytics. Testing scenarios are designed to simulate realistic disposal behavior, and user interaction is limited to RFID scanning and mobile app monitoring.

II. LITERATURE REVIEW

The rapid advancements in IoT (Internet of Things) technology have paved the way for innovative solutions in waste management. Existing studies explore the significance of waste management in various aspects and highlight the potential of IoT-based solutions, such as LoRa communication and deep learning with TensorFlow, to enhance efficiency and sustainability. These efforts aim to improve waste management practices by addressing current challenges. By examining the strengths, limitations, and existing statistics, this review emphasizes the need for a system to reduce the organic waste generated by residents.

A. Study on Waste Generation and Composition in Rapid Residential Development of Suburban Areas in the Kuala Selangor District, Selangor

The literature review focuses on analysing waste generation and composition in the rapidly developing suburban area of Puncak Alam, Kuala Selangor (Siti Rohana Mohd Yatim, 2019). Municipal solid waste (MSW) is identified as a growing concern in Malaysia, accounting for 65% of total waste generated. Over 50% of this comprises food waste, typically disposed of in poorly managed landfills. These unsustainable practices contribute to severe environmental problems, including groundwater contamination, foul odours, and greenhouse gas emissions caused by the degradation of organic matter.



Fig. 1. Research methodology flow

In the study, 200 households were selected through stratified sampling to assess waste generation rates and composition. Waste was collected and weighed daily over four weeks, with materials sorted into categories such as food waste, plastic, paper, metal, and glass. The results show an average per capita waste generation of 1.0 kg/day and approximately 2.5 kg/day per household. Organic waste made

up 65% of the total, followed by paper (8%), plastic (9%), glass (3%), and metal (4%). The methodology's structured approach enables reliable data analysis, and the emphasis on food waste underscores its critical role in landfill challenges. Incorporating socio-economic variables like income level and household size could have strengthened the analysis, but the data still suggest composting as a viable strategy to reduce environmental hazards.

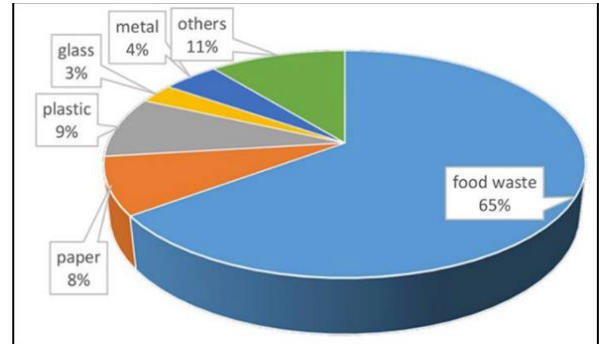


Fig. 2. Percentage of waste composition from the study area

The study emphasizes the inefficiencies in current waste management practices, including the prevalence of open dumpsites and a lack of infrastructure for source separation and recycling. Recommendations for sustainable waste management in the study area include the promotion of recycling programs, the introduction of composting for organic waste, and the adoption of anaerobic digestion technologies to reduce greenhouse gas emissions and conserve landfill space. These strategies aim to address the challenges posed by increasing population and urbanization, which contribute to rising waste generation and environmental degradation.

B. An Internet of Things Based Smart Waste Management System Using LoRa and TensorFlow Deep Learning Model

The paper (Sheng et al., 2020) proposes a smart waste management system integrating IoT and artificial intelligence to optimize waste handling processes. The design includes a bin equipped with a camera for real-time object detection and ultrasonic sensors to monitor fill levels. A Raspberry Pi serves as the main processing unit, controlling waste segregation through servo motors and directing waste into designated compartments for metal, plastic, paper, and general waste. Additional features, such as RFID modules for personnel identification and GPS for location tracking, enhance functionality. The system operates autonomously, reducing manual intervention while ensuring efficient waste separation.

LoRa technology is a key component, enabling long-range, low-power communication for transmitting bin status and location to a central server. The TensorFlow framework, integrated with a lightweight SSD MobileNetV2 model, performs object detection and classification directly on the Raspberry Pi. This approach minimizes data transmission latency by processing waste images locally. The combination of LoRa's communication efficiency and TensorFlow's real-time detection capability allows the system to manage waste effectively, even in remote or large urban areas.



Fig. 3. LoRa with wasp mote and multiprotocol radio shield

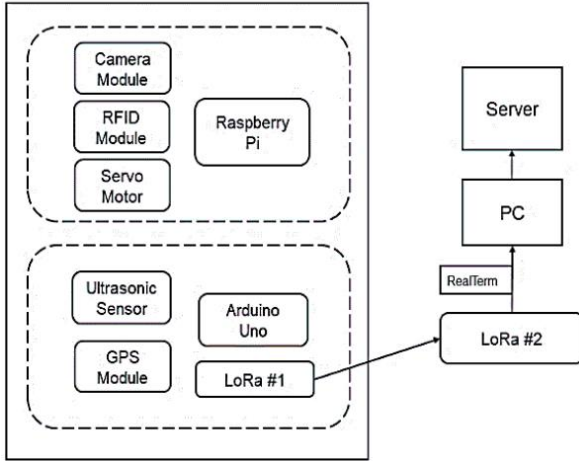


Fig. 4. Block Diagram representing the overall system

Despite its innovations, the system has limitations. The accuracy of the object detection model is constrained by the size and diversity of the training dataset, affecting its ability to classify certain waste types. Additionally, the system is designed to handle only one piece of waste at a time, which may reduce efficiency in high-traffic areas. Expanding the training dataset, increasing model training iterations, and developing solutions to handle simultaneous waste inputs could further enhance the system's capabilities. These improvements would increase the model's robustness and adaptability to real-world scenarios.

C. IoT Based Smart Garbage System for Efficient Food Waste Management

This paper in (Hong et al., 2014) proposed an IoT-based smart garbage system (SGS) designed to manage food waste efficiently through interconnected smart garbage bins (SGBs), routers, and servers. The SGBs operate on battery power for mobility and are integrated into a wireless mesh network for reliable communication. The system includes two domains: a service domain, where residents interact with SGBs to dispose of food waste, and an administration domain, where data is processed, analyzed, and managed.

An adaptive user-oriented charge policy is implemented to motivate residents to reduce food waste. The policy adjusts fees based on monthly reductions in food waste, encouraging sustainable practices. The SGS also features an energy-efficient operation, employing techniques such as sleep mode for SGBs and cooperative communication between bins to optimize battery life.

To manage system resources efficiently, the smart garbage bin (SGB) estimates how long its battery will last using the formula below:

$$d = \frac{P_B}{((\sum_{i=1}^7 P_c(i) + P_w(i)) / 7)} \quad (1)$$

In this formula:

- P_b is the current battery charge level.
- $P_c(i)$ is the power used for communication on day i .
- $P_w(i)$ is the power used for the device's daily operations on day i .
- The average power usage over 7 days is used to estimate how many more days (d) the battery can last.

Using this estimated battery life (d), the system determines which SGB (for example, bin A, B, or C) has the most battery remaining and designates it as the **HSGB** (Head Smart Garbage Bin). This bin will then take the lead in managing communication and coordination for better energy efficiency. Figure 6 illustrates how this energy-efficient cooperation works in the system.

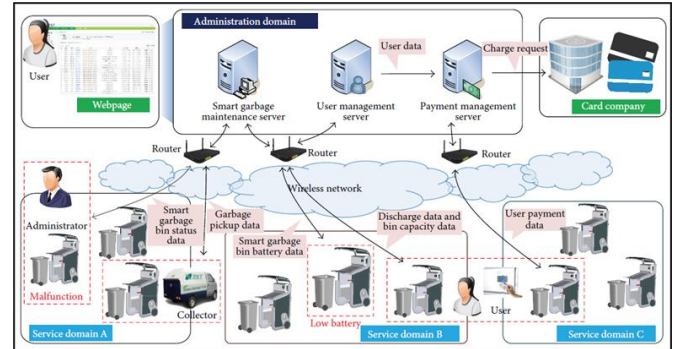


Fig. 5. Overview of smart garbage system

The system was piloted in the Gangnam district, Seoul, for one year. Results showed a 33% reduction in food waste and a 16% improvement in energy efficiency. The hardware includes RFID readers for user authentication, load cells for weight measurement, and a web-based platform for monitoring and managing waste disposal activities. The study concluded that the SGS not only reduces food waste but also provides a scalable, sustainable waste management solution suitable for urban environments.



Fig. 6. Smart garbage bin, LCD, keypad and RFID reader system

III. METHODOLOGY

The Waterfall methodology was selected for this project due to its structured and linear approach, which aligns effectively with the project's requirements. This methodology organizes the work into five phases; including requirement analysis, design, implementation, testing, and maintenance, ensuring that each phase is completed before progressing to the next.

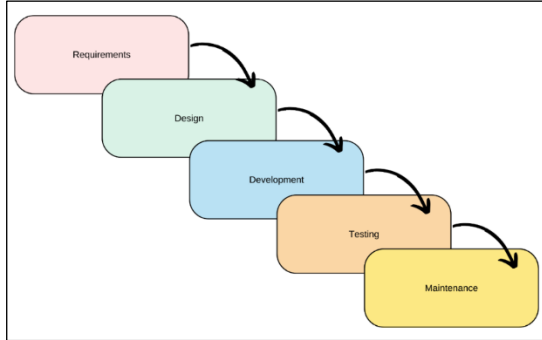


Fig. 7. Waterfall methodology model

A. Hardware and Software Implementation

The development of this project consists of two main parts: hardware and software. Designing the circuit and selecting the appropriate components are crucial to ensuring the system functions effectively. The main components used in this system are:

- i. ESP32 Microcontroller
- ii. Ultrasonic Sensor
- iii. MQ135 Gas Sensor
- iv. RFID Module (NFC V3)

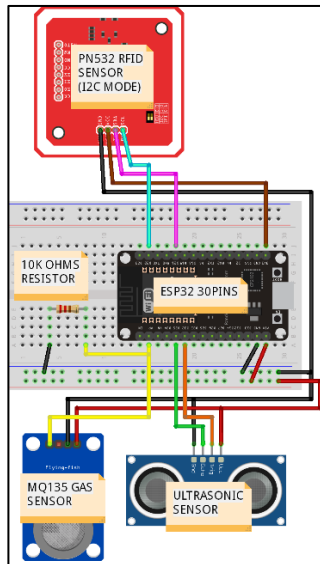


Fig. 8. Circuit diagram

The prototype shown in the report is assembled in an open and loosely connected manner to ensure that each component is clearly visible for documentation and demonstration purposes. This layout allows for easier testing, adjustment, and verification of individual modules, such as

the ESP32, ultrasonic sensor, MQ135 gas sensor, and RFID module. While the current setup is not enclosed, it is intended to be integrated into a physical model of a smart bin in the final stage. The open assembly in the prototype images helps illustrate the wiring and connection logic more transparently before the components are housed in their permanent enclosure.

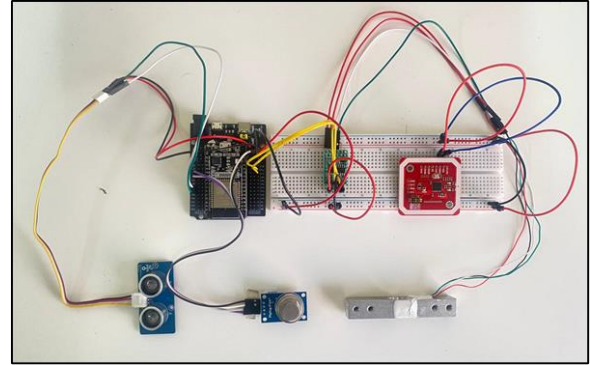


Fig. 9. Project prototype

To program the ESP32 microcontroller and manage sensor interactions, the Arduino IDE was used due to its compatibility with embedded C++ and wide support for IoT devices. Sensor data was transmitted and stored using Firebase, which provided a real-time cloud database for efficient data management and synchronization. For the user-facing interface, a mobile application was developed using Flutter, enabling real-time monitoring of system data through a clean and responsive layout. Visual Studio Code served as the primary development environment, offering robust tools and extensions to streamline both embedded firmware development and cross-platform mobile app coding. List of software used:

- i. Arduino IDE
- ii. Visual Studio Code
- iii. Firebase
- iv. Flutter

B. Project Design

The system architecture is centered around the NodeMCU ESP32, which manages data collection and communication via its built-in Wi-Fi module. It connects to an ultrasonic sensor for bin fill level detection, an MQ135 sensor for monitoring air quality, and an RFID module for user identification. Data from these components is transmitted to a Firebase cloud database, enabling real-time storage and access. A Flutter-based mobile application interfaces with the Firebase database, allowing users to remotely monitor and manage the system. This integration ensures efficient data flow, real-time updates, and effective waste management.

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To calculate the distance measured by the ultrasonic sensor, the system uses the standard formula:

$$Distance(cm) = \frac{Duration \times 0.034}{2} \quad (2)$$

This formula is derived from the speed of sound in air, which is approximately 340 m/s or 0.034 cm/μs. The duration represents the time taken for the ultrasonic pulse to travel to the object and reflect back to the sensor. Dividing by 2 accounts for the round-trip travel time. To ensure consistent and reliable readings, the calculated distance is clamped within a practical range. Any value below 0 cm or above 6.43 cm is reset to the maximum threshold of 6.43 cm, as this reflects the maximum measurable height of the bin in the prototype configuration.

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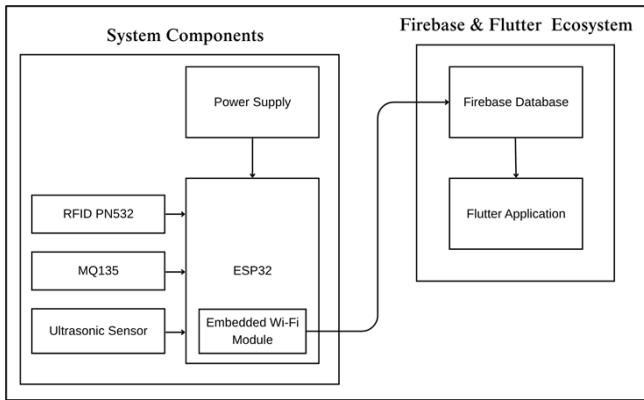


Fig. 10. Block diagram of the project

The system begins by initializing the ESP32's Wi-Fi module and establishing a network connection. If the connection fails, the system continues attempting until successful. Once connected, users are required to scan their RFID cards for authentication; failed scans prompt reattempts until access is granted.

```
// Firestore UUIDs
String uid_user1 = "Amru"; // uid 1
String uid_user2 = "Dalia"; // uid 2
String uid_user3 = "Muuna"; // uid 3
String uid_admin = "jCHE4mkF9XTeSxea7m5UhBlojh1"; // uid admin

String getFirestoreUID(String scannedUID) {
  if (scannedUID == "538F4716") return uid_user1;
  if (scannedUID == "36C54B00") return uid_user2;
  if (scannedUID == "D364AF0E") return uid_user3;
  return "unknown"; // Default fallback instead of "unknown"
}
```

Fig. 11. Pre-registered user

For user authentication, the system relies on a pre-registration approach. Each authorized user's RFID tag is associated with a unique identifier (UID) that is hardcoded into the Arduino firmware. When an RFID tag is scanned, the system compares the scanned UID with a list of predefined

values stored in the code. If a match is found, the corresponding username or user ID is returned and used to identify the user within the Firebase database. This method ensures that only recognized users, whose UIDs have been explicitly defined in the code, can interact with the system. For instance, users such as "Amru," "Dalia," and "Muuna" are assigned specific RFID tag UIDs that the system can identify.

After successful authentication, the user deposits organic waste into the bin. The ultrasonic sensor measures the fill level of the bin, and this data is uploaded to the Firebase database, contributing to the user's monthly waste total. Simultaneously, the MQ135 air quality sensor collects gas concentration data, which is also stored in Firebase. If thresholds are breached, a push notification is sent to the management. Otherwise, all data is displayed on the Flutter-based mobile application, enabling real-time system monitoring and efficient tracking of user-specific waste contributions.

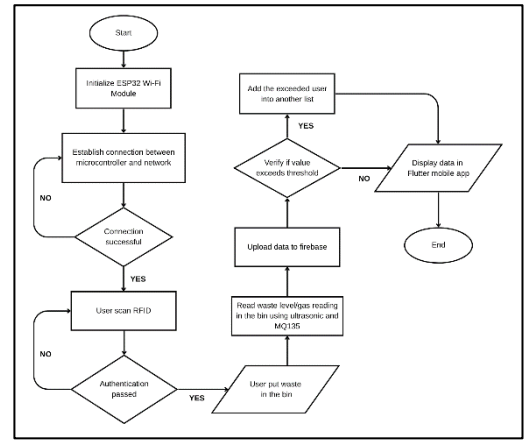


Fig. 12. Flowchart of the system

Before developing the mobile application using Flutter, the app interface was first designed using Canva to enhance clarity and layout visualization. This preliminary design step facilitated a smoother and more efficient coding process later in Visual Studio Code.

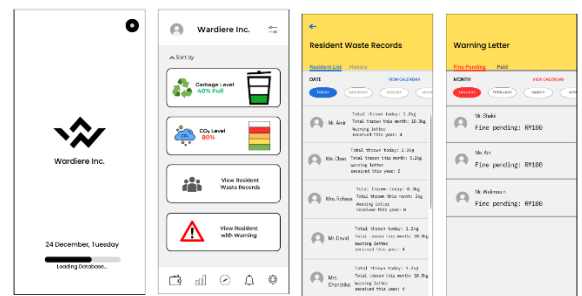


Fig. 13. Early design of the application layout in Canva

C. Data collection

The system collects data from various sensors—such as the ultrasonic sensor for measuring waste bin level and the MQ-135 gas sensor for air quality. The data is read in real time and processed in the ESP32 microcontroller. Once

collected, the data is formatted into JSON objects using the ArduinoJson library and then sent to Firebase using HTTP requests. For example, in the function `pushToRTDB(String uid)`, multiple sensor readings and timestamps are compiled into a structured JSON document:

```
// Prepare JSON data
StaticJsonDocument<400> doc;
doc["rfid_first_scan"] = awaitingSecondScan;
doc["rfid_second_scan"] = !awaitingSecondScan;
doc["final_height_cm"] = String(secondDistance, 2);
doc["final_height_percent"] = secondFillLevel;
doc["initial_height_cm"] = String(firstDistance, 2);
doc["initial_height_percent"] = firstFillLevel;
doc["waste_thrown_cm"] = String(distanceDifferent, 2);
doc["waste_thrown_percent"] = fillLevelDifferent;
doc["timestamp"] = getTimeISO8601();

String jsonPayload;
serializeJson(doc, jsonPayload);

bool success = sendHttpRequest(url, jsonPayload, "PUT");
```

Fig. 14. Sending data for the realtime database

This JSON payload is then uploaded to Firebase Realtime Database (RTDB) via an HTTP PUT request. Similarly, air quality data from the gas sensor is sent to Firestore, as seen in the `pushToGasSubCollection(int ppmGas)` function:

```
void pushToGasSubCollection(int ppmGas) {
    StaticJsonDocument<200> doc;
    doc["fields"]["gas_reading"]["integerValue"] = ppmGas;
    doc["fields"]["timestamp"]["stringValue"] = getTimeISO8601();
}
```

Fig. 15. Sending data for the firebase database

The use of `sendHttpRequest(...)` ensures that the structured JSON is transmitted correctly to the designated Firebase endpoint. These collected data entries are stored in JSON format in both Firebase RTDB and Firestore. This not only makes the data human-readable but also allows it to be easily downloaded or integrated into analytics platforms.

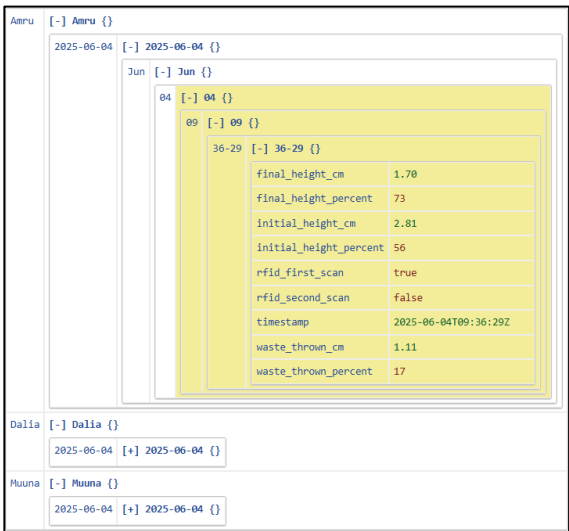


Fig. 16. The downloaded json file viewed in gridview

IV. RESULT & DISCUSSION

The developed system successfully integrated hardware and software components to collect, process, and display real-time sensor data related to impact and movement during Taekwondo practice. Force-sensitive resistor (FSR) sensors captured the impact force values, while the MPU6050 sensor recorded three-dimensional accelerometer and gyroscope readings. These raw values were transmitted via the ESP32 microcontroller to a local MySQL database using a Wi-Fi network.



Fig. 17. Data stored in Firebase realtime database

Figure 17 illustrates how the Firebase Realtime Database (RTDB) stores user interaction data in a structured JSON format. The records are organized by user ID, date, and timestamp, allowing each transaction to be uniquely identified. Each log includes the initial and final height of the waste in both centimeters and percentage, as well as calculated waste thrown and RFID scan flags. This data confirms that the ultrasonic sensor accurately detects changes in waste levels and that the RFID system successfully verifies user identity before allowing interaction. The storage format supports traceability and ease of data analysis for evaluating system performance over time.

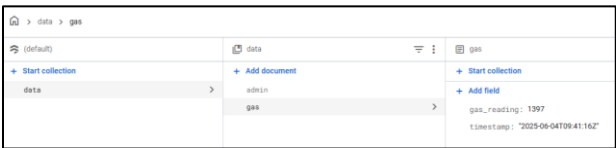


Fig. 18. Data stored in Firestore database

Figure 18 demonstrates how gas sensor readings are logged in Firestore under a specific collection. The values are stored alongside timestamps in ISO 8601 format, enabling precise environmental monitoring. These readings reflect the air quality near the waste bin, captured by the MQ135 gas sensor. The database successfully logs values in real time, confirming that the averaging algorithm and sensor interfacing function correctly. This additional layer of data can help identify when high levels of waste are accompanied by poor air quality, potentially triggering administrative alerts or maintenance actions.

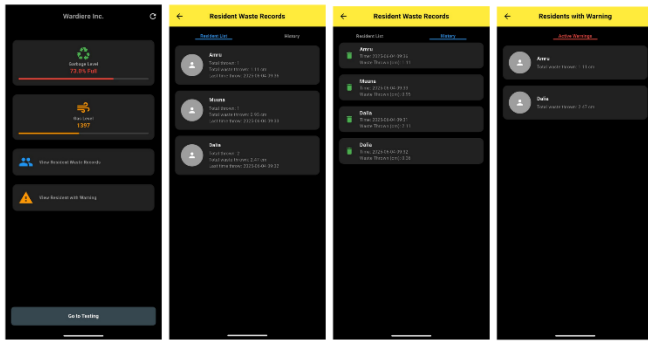


Fig. 19. Final outcome of the mobile application

The final version of the mobile application was developed using Flutter and successfully integrates with Firebase services. It allows administrators to view recorded data remotely, offering a centralized interface for monitoring bin usage and environmental conditions. The real-time functionality and data retrieval capabilities demonstrate the effectiveness of the system architecture. This result confirms that the end-to-end integration—from sensors and microcontroller to cloud and user interface—can operate cohesively in a practical smart bin application.

V. CONCLUSION

This project successfully demonstrated the design and implementation of an IoT-enabled Smart Waste Management System tailored for residential buildings. The system integrated ultrasonic, gas, and RFID sensors with an ESP32 microcontroller and Firebase cloud infrastructure to track and log waste disposal activities. Data such as bin fill levels, gas readings, and user authentication via RFID were collected and stored in structured JSON format within Firebase Realtime Database and Firestore. A mobile application was developed specifically for the building management to monitor resident activity, review waste metrics, and assess bin status remotely.

Despite its effective performance in a prototype environment, several limitations were identified. These include the reliance on pre-registered RFID tags, potential inaccuracies in bin level detection using ultrasonic sensors,

and the need for consistent Wi-Fi connectivity. Furthermore, the system currently lacks a direct interface for residents to receive real-time feedback at the disposal point.

Future improvements will focus on enhancing accuracy and interactivity. Replacing the ultrasonic sensor with a load cell will provide more consistent and precise measurements based on actual waste weight. Additionally, incorporating a display screen on the bin itself will allow residents to view the weight of their disposed waste in real time, encouraging greater environmental accountability. These upgrades, combined with potential expansions in user registration and offline data caching, would make the system more robust, scalable, and suitable for full-scale deployment in residential complexes.

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